Chapter 4

Limit, Continuity, Differentiability and Ordinary Differentiation

- ✓ **Lecture Content:**
  - ✓ 2.2 (Page 83): Limit of a function.
  - ✓ 2.2 (Page 87): One sided limit.
  - ✓ 2.2 (Page 88): Example 7.
  - ✓ 2.2 (Page 90): Vertical Asymptote.

Sample Exercise Set 4.1

1. Sketch and find limit of the function at $x = 1$

$$f(x) = \begin{cases} x^2;\ 0 < x < 1 \\ x; 1 < x < 2 \end{cases}$$

2. Sketch and find limit of the function at $x = -1$ and $x = 1$

$$f(x) = \begin{cases} 1 + x; x < -1 \\ x^2;\ -1 \le x < 1 \\ 2 - x^2; x \ge 1 \end{cases}$$

3. Sketch and find limit at $x = 0$ and $x = \pi$ of

$$f(x) = \begin{cases} 1 + \sin x; x < 0 \\ \cos x; 0 \le x \le \pi \\ \sin x; x > \pi \end{cases}$$

- ✓ **Lecture Content:**
  - ✓ 2.3 (Page 95): Calculating limits using the limit laws.
  - ✓ 2.3 (Page 97-101): Example 2- Example 10.

Sample Exercise Set 4.2

Exercise 2.3 (Page-103) 47 & 50.

- ✓ **Lecture Content:**
  - ✓ 2.5 (Page 114-115): Continuity- just definition.
  - ✓ 2.5 (Page 115): Example 2.

Sample Exercise Set 4.3

Exercise 2.5 (Page 124) 17 & 18.

Exercise 2.5 (Page 125) 39 & 40.

- ✓ **Lecture Content:**
  - ✓ **Differentiability of a function:**

Let the function $f(x)$ be defined on an open interval I and $x_0 \in I$. The function $f(x)$ is said to be differentiable at x_0 if

$$\lim_{\Delta x \to 0} \frac{f(x_0+\Delta x)-f(x_0)}{\Delta x}$$

exists (which is called the 1st principal of differentiation), i.e., **left hand derivative**

$Lf'(x_0) = \lim_{\Delta x \to 0-} \frac{f(x_0+\Delta x)-f(x_0)}{\Delta x}$ is equal to **right hand derivative**

$Rf'(x_0) = \lim_{\Delta x \to 0+} \frac{f(x_0+\Delta x)-f(x_0)}{\Delta x}$.

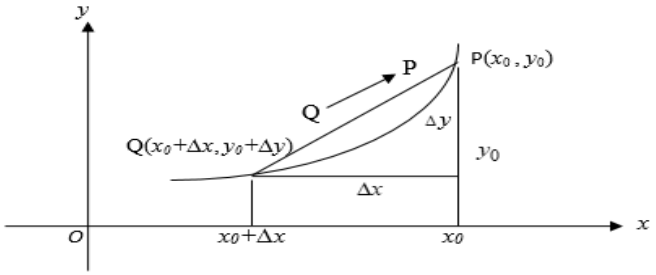


Fig-1: (left hand derivative)

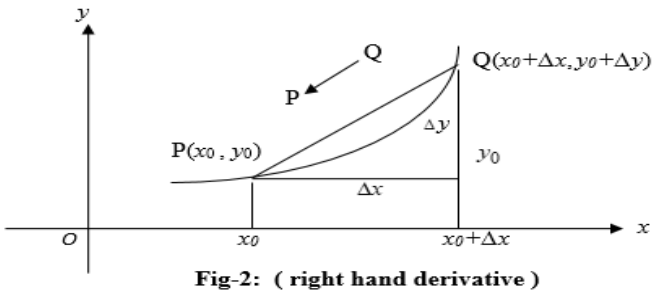


Fig-2: (right hand derivative)

Note: Every differentiable function is continuous but reverse is not always true.

- ✓ **Lecture Content:**
  - ✓ **3.1 (Page 172-179)** Derivatives of polynomial and exponential function.
  - ✓ **3.2 (Page 183-187)** The product and quotient rule.
  - ✓ **3.3 (Page 190-195)** Derivatives of trigonometric functions.
  - ✓ **3.4 (Page 197-204)** The chain rule.
  - ✓ **3.5 (Page 208-213)** Implicit differentiation.
  - ✓ **3.5 (Page 213-214)** Derivatives of inverse trigonometric functions.
  - ✓ **3.6 (Page 218-222)** Derivatives of logarithmic functions.

Sample Exercise Set 4.4

1. Using 1st principal of differentiation find $\frac{d}{dx}(\cos x), \frac{d}{dx}(x^n), \frac{d}{dx}(\sin x)$ and $\frac{d}{dx}(e^{mx})$, $\frac{d}{dx}(\tan x), \frac{d}{dx}(x^2)$.

2. Find $\frac{dy}{dx}$ from the following functions:

(a) **Exercise 3.1 (Page 180):** 7, 11, 23, 16, 18, 20, 30, 32.

(b) **Exercise 3.2 (Page 188):** 5, 11, 12, 17, 19, 25, 26, 20.

(c) **Exercise 3.3 (Page 196):** 1, 3, 2, 5, 8 9, 10, 12, 16.

(d) **Exercise 3.4 (Page 204):** 1-6, 26, 16, 14, 10, 45, 41, 38, 44.

(e) **Exercise 3.5 (Page 214):** 5, 8, 10, 11, 12, 17, 49, 51, 59, 60.

(f) **Exercise 3.6 (Page 222):** 2, 3, 4, 7, 9, 19, 21, 24.

- ✓ **Parametric Differentiation**

Suppose that a curve is given by the parametric equation $x = f(t)$ and $y = g(t)$,where $f(t)$ and $g(t)$ are both differentiable. Then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{g'(t)}{f'(t)}$$

One of the parametric equations of conic are:

1. Parametric equation of circle: $x = r\cos t$ and $y = r\sin t$, where $r > 0, 0 \le t \le \pi$.
2. Parametric equation of ellipse: $x = a\cos t$ and $y = b\sin t$, where $a, b > 0, 0 \le t \le 2\pi$.
3. Parametric equation of parabola: $x = t$ and $y = t^2$.
4. Parametric equation of hyperbola: $x = a\sec t$ and $y = b\tan t$.

Note: Parametric equations of a curve is not unique.

Example: Let $x = a(\theta - \sin\theta)$ and $y = a(1-\cos\theta)$. Then

$$\frac{dx}{d\theta} = a(1-\cos\theta) \text{ and } \frac{dy}{d\theta} = a\sin\theta$$

and hence $$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\sin\theta}{1-\cos\theta}.$$

Sample Exercise Set 4.5

1. Find $\frac{dy}{dx}$ in terms of the parameter *t*.

(a) $x = t^4 + 1, y = t(t^2 + 3)$, (b) $x = \sin t, y = \exp(2t)$, (c) $x = t^2 - \cos 3t, y = \sin 2t$,

(d) $x = 5\sin^3 t$, $y = 4\cos t$, and (e) $x = a(1+\sin 2t)$, $y = a(t - \cos 3t)$.

- ✓ **Lecture Content**
  - ✓ **2.7 (Page 140-148):** Derivatives and Rates of Change.

Example: Find the equations of the tangent and normal lines to the curve $y = x^2$ at $(1,1)$.

Solution: $\frac{dy}{dx} = 2x$. When $x = 1$, $\frac{dy}{dx} = 2$

Equation of tangent line at the point $(1,1)$ is

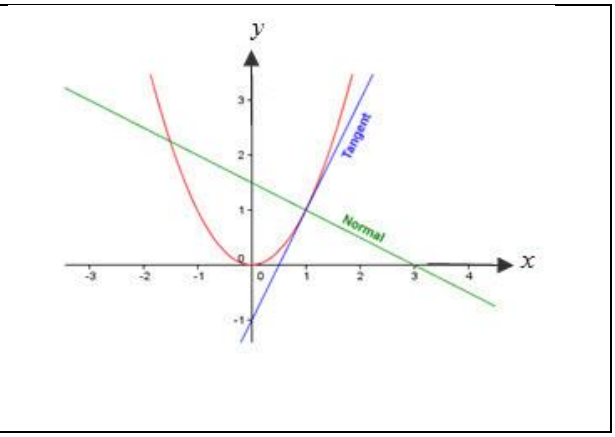
$$y - 1 = 2(x - 1)$$

Or, $2x - y = 1$

Equation of normal line at the point $(1,1)$ is

$$y - 1 = \frac{-1}{2}(x - 1)$$

Or, $x + 2y = 3$



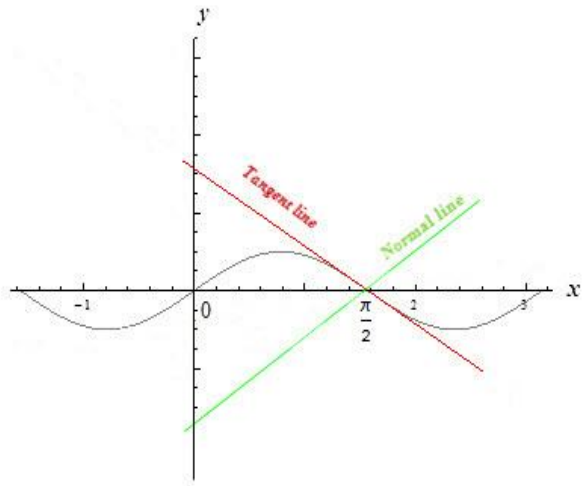
Example: Find the equations of the tangent and normal lines to the curve $y = \sin 2x$ at $\left(\frac{\pi}{2}, 0\right)$.

Solution: $\frac{dy}{dx} = 2\cos 2x$. When $x = \frac{\pi}{2}$, $\frac{dy}{dx} = -2$.

Equation of tangent line at the point $\left(\frac{\pi}{2}, 0\right)$ is: $y - 0 = -2\left(x - \frac{\pi}{2}\right)$

Or, $2x + y = \pi$

Equation of normal line at the point $\left(\frac{\pi}{2}, 0\right)$ is: $y - 0 = \frac{1}{2}\left(x - \frac{\pi}{2}\right)$ Or, $x - 2y = \frac{\pi}{2}$.



Example: Find the equations of the tangent and normal lines to the curve $y = \exp(x)$ at the point where $x = 0$.

Solution: At $x=0,\ y=\exp(0)=1.$

Now, $\dfrac{dy}{dx}=\exp(x).$ When $x=0,$ $\dfrac{dy}{dx}=1$

Equation of tangent line at the point $(0,1)$ is

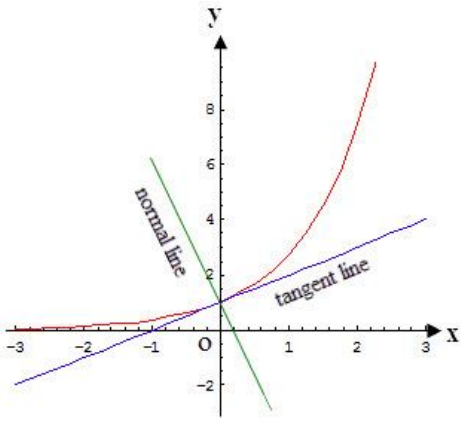
$$y-1=1(x-0)$$

Or, $y=x+1$

Equation of normal line at the point $(0,1)$ is

$$y-1=-\frac{1}{1}(x-0)$$

Or, $y=-x+1.$



Example: Find the equations of the tangent and normal lines to the curve $y=\ln x$ at the point where $x=1.$

Solution: At $x=1,\ y=\ln 1=0.$

Now, $\dfrac{dy}{dx}=\dfrac{1}{x}.$ When $x=1,$ $\dfrac{dy}{dx}=1$

Equation of tangent line at the point $(1,0)$ is

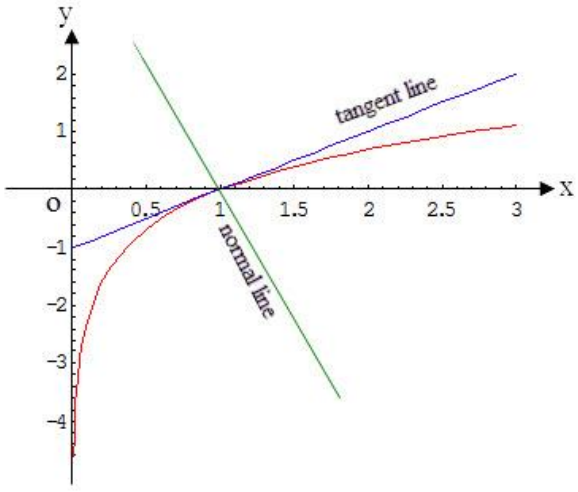
$$y-0=1(x-1)$$

Or, $y=x-1$

Equation of normal line at the point $(1,0)$ is

$$y-0=-\frac{1}{1}(x-1)$$

Or, $y=-x+1.$



Example:

If a free- falling object is dropped from a height of 100 feet, and air resistance is neglected, the height h (in feet) of the object at a time t (in seconds) is given by $h=-16t^2+100.$

Find the average velocity of the object over each time interval.

a) [1,2] b) [1, 1.5] c) [1,1.1]

Solution: You can use the position $h=-16t^2+100$ to determine the heights at $t=1$, $t=1.1$, $t=1.5$ and $t=2$ as shown in the table.

t (in seconds)	0	1	1.1	1.5	2
h(in feet)	100	84	80.64	64	36

a) For the interval [1,2], the object fall from a height of 84 feet to a height of 36 feet. So, the average velocity is

$$\frac{\Delta h}{\Delta t} = \frac{36-84}{2-1} = \frac{-48}{1} = -48 \text{ feet per second.}$$

b) For the interval [1, 1.5], the average velocity is

$$\frac{\Delta h}{\Delta t} = \frac{64-84}{1.5-1} = \frac{-20}{0.5} = -40 \text{ feet per second.}$$

c) For the interval [1,1.1], the average velocity is

$$\frac{\Delta h}{\Delta t} = \frac{80.64-84}{1.1-1} = \frac{-3.36}{0.1} = -33.6 \text{ feet per second.}$$

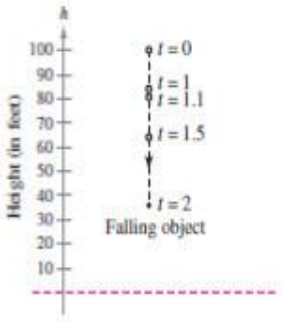


Figure: Some falling objects have considerable air resistance. Other falling objects have negligible air resistance. When modeling a falling-body problem, you must decide whether to account for air resistance or neglect it.

Sample Exercise Set 4.7

(a) **Exercise 2.7 (Page 148):** 5-8.

(b) **Example 3 (Page 143)**.

(c) **Example 5 (Page 145)**.

✓ L'Hôpital Rule : (Indeterminate forms)

✓ Indeterminate forms:

A mathematical expression can also be said to be indeterminate if it is not definitively or precisely determined. Certain forms of limits are said to be indeterminate when merely knowing the limiting behavior of individual parts of the expression is not sufficient to actually determine the overall limit.

For example, a limit of the form $\frac{0}{0}$, i.e., $\lim_{x\to a}\frac{f(x)}{g(x)}$ where $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$, is indeterminate since the value of the overall limit actually depends on the limiting behavior of the combination of the two functions (e.g., $\lim_{x\to 0}\frac{x}{x} = 1$, while $\lim_{x\to 0}\frac{x^2}{x} = 0$).

There are seven indeterminate forms involving 0, 1 and ∞:

$$\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, \; 0 \times \infty, \; 0^0, \infty^0 \text{ and } 1^\infty$$

for $\frac{0}{0}$ indeterminate form:

Suppose that *f* and *g* are differentiable functions on an open interval containing $x = a$, except possibly at $x = a$, and that $\lim_{x \to a} f(x) = 0$ and $\lim_{x \to a} g(x) = 0$.

If $\lim_{x \to a}\left[\frac{f'(x)}{g'(x)}\right]$ has a finite limit, or if this is $+\infty$ or $-\infty$, then $\lim_{x \to a}\left[\frac{f(x)}{g(x)}\right] = \lim_{x \to a}\left[\frac{f'(x)}{g'(x)}\right]$.

Example: Evaluate $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$.

Solution: $\lim_{x \to 0} \frac{1 - \cos x}{x^2} \qquad \left[\frac{0}{0} \text{ form}\right]$

$= \lim_{x \to 0} \frac{\sin x}{2x} \qquad \left[\frac{0}{0} \text{ form}\right]$ [using L'Hôpital rule]

$= \lim_{x \to 0} \frac{\cos x}{2} = \frac{1}{2}.$ [using L'Hôpital rule]

Example: Evaluate $\lim_{x \to 0^-} \frac{\tan x}{x^2}$.

Solution: $\lim_{x \to 0^-} \frac{\tan x}{x^2} \qquad \left[\frac{0}{0} \text{form}\right]$

$= \lim_{x \to 0^-} \frac{\sec^2 x}{2x} = -\infty.$ [using L'Hôpital rule]

for $\frac{\infty}{\infty}$ indeterminate form:

Suppose that f and g are differentiable functions on an open interval containing $x = a$, except possibly at $x = a$, and that $\lim_{x \to a} f(x) = \infty$ and $\lim_{x \to a} g(x) = \infty$.

If $\lim_{x\to a}\left[\frac{f'(x)}{g'(x)}\right]$ has a finite limit, or if this is $+\infty$ or $-\infty$, then $\lim_{x\to a}\left[\frac{f(x)}{g(x)}\right]=\lim_{x\to a}\left[\frac{f'(x)}{g'(x)}\right]$

Example: Evaluate $\lim_{x\to+\infty}\frac{x}{e^x}$.

Solution: $\lim_{x\to+\infty}\frac{x}{e^x}$ $\left[\frac{\infty}{\infty}\text{form}\right]$

$=\lim_{x\to+\infty}\frac{1}{e^x}=0.$ [Using L'Hôpital rule]

Note: Indeterminate forms other than $\frac{0}{0}$ and $\frac{\infty}{\infty}$, should be reduced to the form either $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

Indeterminate form of $0\times\infty$ or $\infty\times 0$:

Example: $\lim_{x\to0} x\ln x$

Solution: $\lim_{x\to0^+} x\ln x$ $[0\times\infty \text{ form}]$

$=\lim_{x\to0^+}\frac{\ln x}{\frac{1}{x}}$ $\left[\frac{-\infty}{\infty}\text{form}\right]$

$=\lim_{x\to0^+}\frac{\frac{1}{x}}{-\frac{1}{x^2}}$ [applying L'Hospital rule]

$=\lim_{x\to0^+}-\frac{1}{\frac{1}{x}}$

$=\lim_{x\to0^+}-x=0$.

Indeterminate form of $\infty-\infty$:

Example: $\lim_{x\to0}\left[\frac{1}{x}-\frac{1}{x^2}\ln(1+x)\right]$

Solution:

$\lim_{x\to0}\left[\frac{1}{x}-\frac{1}{x^2}\ln(1+x)\right]$ $[\infty-\infty \text{ form}]$

$=\lim_{x\to0}\frac{x-\ln(1+x)}{x^2}$ $\left[\frac{0}{0}\text{ form}\right]$

$=\lim_{x\to0}\frac{1-\frac{1}{1+x}}{2x}$ [applying L'Hospital rule]

$= \lim_{x\to 0} \frac{1}{2(1+x)} = \frac{1}{2}$.

Indeterminate form of 0^0:

Example: $\lim_{x\to 0^+} x^{3x}$

Solution: Let, $u = \lim_{x\to 0} x^{3x}$ $[0^0 \text{form}]$

$\Rightarrow \ln u = \lim_{x\to 0^+} \ln x^{3x}$

$\Rightarrow \ln u = \lim_{x\to 0^+} 3x \ln x$ $[0 \times (-\infty) \text{ form}]$

$\Rightarrow \ln u = \lim_{x\to 0^+} \frac{3\ln x}{\frac{1}{x}}$ $\left[\frac{\infty}{\infty} \text{ form}\right]$

$\Longrightarrow \ln u = \lim_{x\to 0^+} \frac{\frac{3}{x}}{-\frac{1}{x^2}}$

$\Longrightarrow \ln u = \lim_{x\to 0^+} (-3x) = 0$

$\Longrightarrow u = e^0 = 1.$

Indeterminate form of 1^∞:

Example: $\lim_{x\to 0} (\cos x)^{\cot x}$

Let, $u = \lim_{x\to 0} (\cos x)^{\cot x}$ $[1^\infty \text{ form}]$

$\Longrightarrow \ln u = \lim_{x\to 0} \cot x \ln(\cos x)$ $[\infty \times 0 \text{ form}]$

$\Longrightarrow \ln u = \lim_{x\to 0} \frac{\ln(\cos x)}{\tan x}$ $\left[\frac{0}{0} \text{ form}\right]$

$\Longrightarrow \ln u = \lim_{x\to 0} \frac{\frac{1}{\cos x}(-\sin x)}{\sec^2 x}$

$\Longrightarrow \ln u = 0$

$\Longrightarrow u = e^0 = 1.$

(Ans.)

Indeterminate form of ∞^0 form:

Example: $\lim_{x\to 0} (\cot x)^{\tan x}$

Let, $u = \lim_{x\to 0} (\cot x)^{\tan x}$ $[\infty^0 \text{ form}]$

$\Longrightarrow \ln u = \lim_{x\to 0} \tan x \ln(\cot x)$ $[0 \times \infty \text{ form}]$

$\Longrightarrow \ln u = \lim_{x\to 0} \frac{\ln(\cot x)}{\cot x}$ $\left[\frac{\infty}{\infty} \text{ form}\right]$

$\Longrightarrow \ln u = \lim_{x\to 0} \frac{\frac{1}{\cot x}(-\csc^2 x)}{-\csc^2 x}$

$\Longrightarrow \ln u = \lim_{x\to 0} \tan x$

$\Longrightarrow u = e^0 = 1.$

✓ **Note:** $0^{\infty}, 0^{-\infty}, \infty^{\infty} and\ \infty^{-\infty}$ are not indeterminate forms.

Sample Exercise set 4.6

1. Evaluate the following limits using **L'Hôpital rule** (if possible).

(a) $\lim_{x\to 1} \frac{x^4-4x^3+6x^2-4x+1}{x^3-3x^2+3x-1}$,

(b) $\lim_{x\to 0} \frac{\ln x^2}{\ln(\sin^2 x)}$,

(c) $\lim_{x\to 0^+} \frac{\ln x}{\csc x}$,

(d) $\lim_{x\to\infty} 2^x \sin\left(\frac{a}{2^x}\right)$,

(e) $\lim_{x\to 0^+} x\ln x$

(f) $\lim_{x\to 0} \left[\frac{1}{x} - \frac{1}{x^2}\ln(1+x)\right]$,

(g) $\lim_{x\to 0} \left(\frac{1}{x^2} - \frac{1}{x\sin x}\right)$,

(h) $\lim_{x\to 0} \left(\frac{1}{x^2} - \frac{1}{x\sin x}\right)$,

(i) $\lim_{x\to 0^+} x^{3x}$

(j) $\lim_{x\to 0}(\cos x)^{\cot x}$,

(k) $\lim_{x\to\infty} \left(1+\frac{1}{x}\right)^x$,

(l) $\lim_{x\to 0} (x+e^x)^{\frac{1}{x}}$,

(m) $\lim_{x\to 0}(1+x+x^2)^{\frac{1}{x}}$,

(n) $\lim_{x\to 0}(\cos x)^{\cot^2 x}$,

(o) $\lim_{x\to\infty} \frac{x^2-3x-3}{4x^2+5}$,

(p) $\lim_{x\to 4^+} \frac{\sqrt{x^2-16}}{x+4}$,

(q) $\lim_{x\to 2^+} \frac{\ln(x-2)}{\ln(x^2-4)}$,

(r) $\lim_{x\to 0}\ (\sin x)^{\sqrt{x}}$,

(s) $\lim_{x\to 2} \frac{x^2-4}{\sqrt{x+2}-\sqrt{3x-2}}$,

(t) $\lim_{x\to 1^-} \sqrt{1-x}\ \ln\left(\ln\left(\frac{1}{x}\right)\right)$.

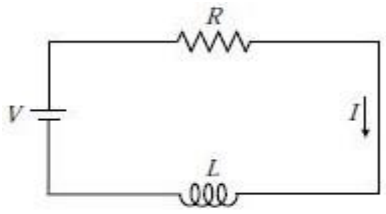
Solution

1. **(a)** 0, **(b)** 1, **(c)** 0, **(d)** a, **(e)** 0, **(f)** $\frac{1}{2}$, **(g)** $-\frac{1}{6}$, **(h)** 1, **(i)** 1, **(j)** e, **(k)** e, **(l)** e^2, **(m)** e, **(n)** $e^{-\frac{1}{2}}$, **(o)** $\frac{1}{4}$, **(p)** 0, **(q)** 1, **(r)** 1, **(s)** -8, **(t)** 1.

2. The accompanying schematic diagram represents an electrical circuit consisting of an electromotive force that produces a voltage V, a resistor with resistance R, and an indicator with inductance L. It is shown in electrical circuit theory that if the voltage is first applied at time $t = 0$, then the current I flowing through the circuit at time t is given by

$$I = \frac{V}{R}\left(1 - e^{-\frac{Rt}{L}}\right)$$

What is the effect on the current at a fixed time t if the resistance approaches zero ($i.e., R \to 0^+$)?



Solution: $\boldsymbol{\frac{Vt}{L}}$